

Tech Mahindra - 100 Technical MCQ Questions

Tech Mahindra

For BTech Computer Science / IT Students

Object-Oriented Programming

1. Which of these is an example of compile-time polymorphism?

- A) Method overriding
- B) Operator overloading (in C++)
- C) Dynamic method dispatch
- D) Runtime type identification

2. What is object composition?

- A) Building complex objects by combining simpler objects
- B) Inheriting from multiple classes
- C) Creating static objects
- D) Deleting objects

3. Which of the following is true regarding static methods and polymorphism?

- A) Static methods support runtime polymorphism
- B) Static methods are resolved at compile time and do not support runtime polymorphism
- C) Static methods cannot be called
- D) Static methods must be abstract

4. A class that cannot be instantiated directly is called a(n):

- A) Abstract class
- B) Final class
- C) Static class
- D) Sealed class

5. In Java, which keyword is used to inherit a class?

- A) implements
- B) extends
- C) inherits
- D) super

6. What is the diamond problem in OOP?

- A) An ambiguity arising from multiple inheritance of the same base class
- B) A memory leak issue
- C) A sorting algorithm's worst case
- D) A type of exception

7. What is method hiding in OOP?

- A) When a static method in a subclass has the same signature as one in the superclass
- B) When a private method is called
- C) When a method throws an exception
- D) When constructors are overloaded

8. Can a constructor be overridden?

- A) Yes, always
- B) No, constructors cannot be overridden
- C) Only static constructors can
- D) Only in interfaces

9. In Python, which method is automatically called when an object is created?

- A) `__new__` then `__init__`
- B) `__del__`
- C) `__call__`
- D) `__str__`

10. Which of the following best describes 'abstraction'?

- A) Showing only relevant details and hiding implementation complexity
- B) Combining data and methods
- C) Creating multiple objects
- D) Executing code faster

11. Which design principle states 'favor composition over inheritance'?

- A) SOLID-related OOP design guideline
- B) A rule that disallows interfaces
- C) A Java-only rule
- D) A database design rule

12. What is a 'friend' function in C++?

- A) A function that has access to private/protected members of a class
- B) A public function only
- C) A function that cannot access class members
- D) A default constructor

13. Which OOP principle is primarily used to reduce code duplication by reusing existing classes?

- A) Inheritance
- B) Polymorphism
- C) Abstraction
- D) Encapsulation

14. Which of these is true about interfaces in Java 8+?

- A) They can have default and static methods with bodies
- B) They can have instance variables with values changed at runtime
- C) They support constructors
- D) They cannot be implemented by multiple classes

15. Which of the following correctly describes 'late binding'?

- A) Method call resolved at compile time
- B) Method call resolved at runtime based on object type
- C) A binding used only for static variables
- D) A binding used for interfaces only

16. What is operator overloading?

- A) Giving additional meaning to an operator for user-defined types
- B) Removing operators from a language
- C) A compilation error
- D) A runtime exception

DBMS & SQL

17. What is a composite key?

- A) A key formed by combining two or more columns to uniquely identify a row
- B) A key that references another table
- C) An automatically generated key
- D) A key with only one column

18. What is a trigger in SQL?

- A) A special stored procedure automatically executed in response to certain events on a table
- B) A type of index
- C) A join operation
- D) A constraint on primary keys

19. What is a view in SQL?

- A) A physical copy of a table
- B) A virtual table based on the result of a SQL query
- C) A type of index
- D) A stored procedure

20. What is the purpose of an index in a database?

- A) To enforce referential integrity
- B) To speed up data retrieval operations
- C) To store backup data
- D) To encrypt table data

21. What is a self-join?

- A) A join of a table with itself
- B) A join between two different databases
- C) A join without any condition
- D) An invalid SQL operation

22. Which index type stores data rows in the same order as the index?

- A) Non-clustered index
- B) Clustered index
- C) Bitmap index
- D) Hash index

23. Which SQL statement is used to combine the result-set of two or more SELECT statements with the same number of columns?

- A) JOIN
- B) UNION
- C) MERGE
- D) GROUP BY

24. Which SQL aggregate function returns the number of rows matching a condition?

- A) SUM()
- B) COUNT()
- C) AVG()
- D) MAX()

25. Which of the following is true about a candidate key?

- A) It can contain NULL values
- B) Every candidate key can be a primary key, but only one is chosen as such
- C) A table can have only one candidate key
- D) It must always be a foreign key

26. Which SQL keyword is used to remove duplicate rows from a result set?

- A) UNIQUE
- B) DISTINCT
- C) FILTER
- D) ONLY

27. What is a schema in a database context?

- A) The logical structure that defines tables, columns, and relationships
- B) A backup file
- C) A stored procedure
- D) A query optimizer

28. What is the purpose of the GROUP BY clause?

- A) To sort data in ascending order
- B) To arrange identical data into groups for aggregate calculations
- C) To filter rows before aggregation
- D) To join multiple tables

29. What is the difference between WHERE and HAVING clauses?

- A) WHERE filters rows before grouping, HAVING filters groups after aggregation
- B) They are identical in function
- C) HAVING is used only with SELECT *
- D) WHERE cannot use comparison operators

30. Which of the following is an example of a NoSQL database?

- A) MySQL
- B) PostgreSQL
- C) MongoDB
- D) Oracle

31. Which of these correctly describes a many-to-many relationship implementation?

- A) Implemented directly with a foreign key in one table
- B) Implemented using a junction/associative table with foreign keys to both tables
- C) Not possible in relational databases
- D) Requires a trigger

32. What does 'sharding' mean in database scaling?

- A) Splitting a large database horizontally across multiple servers
- B) Encrypting data
- C) Creating a backup
- D) Normalizing tables

33. Which normal form eliminates transitive dependency?

- A) 1NF
- B) 2NF
- C) 3NF
- D) 0NF

34. What does the term 'referential integrity' ensure?

- A) That foreign key values always match a primary key value in the referenced table (or are NULL)
- B) That all columns are indexed
- C) That all queries run faster
- D) That tables cannot be dropped

35. Which of the following best defines a stored procedure?

- A) A precompiled collection of SQL statements stored in the database and executed as a unit
- B) A type of foreign key
- C) A column constraint
- D) A backup mechanism

36. What is the main advantage of using an ORM (Object-Relational Mapping) tool?

- A) It eliminates the need for a database entirely
- B) It lets developers interact with the database using objects instead of raw SQL
- C) It automatically normalizes data
- D) It replaces indexes

37. Which isolation level allows 'dirty reads'?

- A) Serializable
- B) Repeatable Read
- C) Read Committed
- D) Read Uncommitted

38. Which SQL function is used to combine strings from multiple rows into one string (varies by RDBMS)?

- A) CONCAT() only
- B) GROUP_CONCAT()/STRING_AGG() depending on RDBMS
- C) MERGE()
- D) JOIN()

Operating Systems

39. Which of the following best defines 'mutual exclusion' as a deadlock condition?

- A) A resource can be used by multiple processes simultaneously
- B) At least one resource must be held in a non-shareable mode
- C) Processes never request resources
- D) Resources are always preemptable

40. What is the difference between a hard link and a symbolic link in a file system?

- A) A hard link points directly to the inode; a symbolic link is a separate file containing a path to the target
- B) They are identical
- C) Symbolic links cannot be broken
- D) Hard links can span across different file systems while symbolic links cannot

41. Which of the following is true about a microkernel architecture?

- A) It moves most services to user space, keeping the kernel minimal
- B) It has no scheduler
- C) It cannot support IPC
- D) It is always slower with no exceptions

42. Which of the following is an example of external fragmentation?

- A) Free memory scattered in small blocks between allocated partitions, none large enough for a new request
- B) Wasted space within an allocated fixed-size block
- C) A page fault
- D) A deadlock

43. Which file system attribute determines the smallest unit of disk space allocation?

- A) Sector/Block size
- B) File descriptor
- C) Inode number
- D) Directory depth

44. What is a monolithic kernel?

- A) A kernel where all OS services run in kernel space as a single large process
- B) A kernel with no drivers
- C) A kernel used only in embedded systems
- D) A kernel that cannot be modified

45. What is the primary function of an operating system?

- A) To compile programs
- B) To manage hardware resources and provide services to applications
- C) To design websites
- D) To encrypt files only

46. Which scheduling algorithm can cause starvation?

- A) Round Robin
- B) Priority Scheduling (without aging)
- C) FCFS
- D) None of these

47. What is the banker's algorithm used for?

- A) Deadlock avoidance by checking safe states before allocating resources
- B) Memory compression
- C) File compression
- D) CPU scheduling

48. What is the role of the dispatcher in CPU scheduling?

- A) To select which process runs next based purely on priority
- B) To give control of the CPU to the process selected by the short-term scheduler
- C) To manage disk I/O only
- D) To handle network requests

49. What is a deadlock?

- A) A state where two or more processes are waiting indefinitely for resources held by each other
- B) A process that has finished execution
- C) A type of memory leak
- D) A CPU scheduling technique

50. Which of the following is NOT one of the four necessary conditions for deadlock?

- A) Mutual Exclusion
- B) Hold and Wait
- C) Preemption
- D) Circular Wait

51. What does the term 'fragmentation' refer to in memory management?

- A) Wasted memory space due to allocation and deallocation patterns (internal or external)
- B) A type of process priority
- C) A CPU scheduling delay
- D) A file system backup issue

52. What is an inode in a Unix-like file system?

- A) A data structure storing metadata about a file (permissions, size, pointers to data blocks) but not its name
- B) The file's name only
- C) A type of directory
- D) A kernel module

53. Which of these is a necessary condition to prevent deadlock by breaking 'circular wait'?

- A) Impose a strict ordering of resource acquisition
- B) Allow unlimited resource requests
- C) Never release resources
- D) Increase the number of processes

54. What is spooling?

- A) Simultaneous Peripheral Operations On-Line - buffering data for devices like printers
- B) A type of deadlock
- C) A page replacement algorithm
- D) A CPU scheduling technique

Computer Networks

55. Which layer of the OSI model is responsible for routing?

- A) Data Link Layer
- B) Network Layer
- C) Transport Layer
- D) Session Layer

56. How many layers are there in the OSI model?

- A) 5
- B) 6
- C) 7
- D) 8

57. What is the primary purpose of a load balancer?

- A) To distribute incoming network traffic across multiple servers to improve reliability and performance
- B) To encrypt data at rest
- C) To act as a firewall only
- D) To assign static IP addresses

58. What is the main difference between TCP and UDP?

- A) TCP is connectionless, UDP is connection-oriented
- B) TCP is connection-oriented and reliable, UDP is connectionless and faster but unreliable
- C) There is no difference
- D) UDP guarantees ordered delivery while TCP does not

59. Which of the following best describes a CDN (Content Delivery Network)?

- A) A distributed network of servers that delivers content to users from a location geographically closer to them
- B) A type of firewall
- C) A DNS resolution protocol
- D) A database replication technique

60. Which protocol operates at the Transport layer and is connection-oriented?

- A) UDP
- B) IP
- C) TCP
- D) ICMP

61. What is a MAC address?

- A) A unique hardware identifier assigned to a network interface
- B) An IP address used only for routing
- C) A type of DNS record
- D) A session identifier

62. What does DHCP stand for?

- A) Dynamic Host Configuration Protocol
- B) Data Host Control Protocol
- C) Direct Host Connection Protocol
- D) Domain Host Configuration Protocol

63. Which protocol is used to resolve a domain name to an IP address?

- A) FTP
- B) DNS
- C) SMTP
- D) ARP

64. Which of the following is an application-layer protocol?

- A) TCP
- B) IP
- C) HTTP
- D) Ethernet

65. What is the default port number for HTTPS?

- A) 80
- B) 21
- C) 443
- D) 22

66. What is the default port number for HTTP?

- A) 21
- B) 25
- C) 80
- D) 443

67. What does a 'ping' command test?

- A) Reachability and round-trip time to a host using ICMP echo requests
- B) DNS resolution only
- C) Bandwidth of a connection
- D) Server uptime history

68. Which protocol is used for secure remote login to a server?

- A) Telnet
- B) SSH
- C) FTP
- D) SMTP

Data Structures & Algorithms

69. What is dynamic programming primarily used for?

- A) Solving problems by breaking them into overlapping subproblems and storing results to avoid recomputation
- B) Sorting arrays only
- C) Random number generation
- D) Compiling code faster

70. Which data structure is typically used to implement BFS?

- A) Stack
- B) Queue
- C) Heap
- D) Array only

71. What is the key property of a min-heap?

- A) The parent node is always greater than its children
- B) The parent node is always smaller than or equal to its children
- C) It must be a complete binary tree only, with no ordering
- D) It is always a balanced BST

72. Which of the following problems is a classic example solved using dynamic programming?

- A) Binary search
- B) 0/1 Knapsack problem
- C) Bubble sort
- D) Linear search

73. Which algorithm is used to find the Minimum Spanning Tree of a graph?

- A) Dijkstra's Algorithm
- B) Kruskal's or Prim's Algorithm
- C) Bellman-Ford Algorithm
- D) Floyd-Warshall Algorithm

74. What is the time complexity of Dijkstra's algorithm using a binary heap?

- A) $O(V^2)$
- B) $O(E \log V)$
- C) $O(V + E)$
- D) $O(E^2)$

75. What is the time complexity of finding the minimum element in an unsorted array?

- A) $O(1)$
- B) $O(\log n)$
- C) $O(n)$
- D) $O(n \log n)$

76. Which data structure would you use to check if a string of brackets is balanced?

- A) Queue
- B) Stack
- C) Array
- D) Linked List

77. What is memoization?

- A) Storing the results of expensive function calls and reusing them when the same inputs occur again
- B) A sorting technique
- C) A type of recursion without a base case
- D) A method to delete duplicate array elements

78. What is the time complexity of Bubble Sort in the worst case?

- A) $O(n)$
- B) $O(n \log n)$
- C) $O(n^2)$
- D) $O(\log n)$

79. Which graph algorithm is used to find the shortest path in a graph with possible negative edge weights (no negative cycles)?

- A) Dijkstra's Algorithm
- B) Bellman-Ford Algorithm
- C) Prim's Algorithm
- D) Kruskal's Algorithm

80. Which data structure is typically used to implement DFS (iteratively)?

- A) Queue
- B) Stack
- C) Heap
- D) Hash table

81. What does BFS stand for in graph traversal?

- A) Best First Search
- B) Breadth First Search
- C) Binary First Search
- D) Backtrack First Search

82. What is a hash collision?

- A) When two different keys map to the same hash bucket/index
- B) When a hash table runs out of memory
- C) When a key is deleted incorrectly
- D) When two hash tables merge

83. What is the average time complexity of searching in a well-implemented hash table?

- A) $O(n)$
- B) $O(\log n)$
- C) $O(1)$
- D) $O(n^2)$

84. Which of the following is a common technique to resolve hash collisions?

- A) Chaining or open addressing
- B) Binary search
- C) Recursion
- D) Memoization

Programming Language Fundamentals

85. What does 'immutable' mean for an object (e.g., a String in Java)?

- A) Its value can be changed after creation
- B) Its state cannot be changed after it is created
- C) It can only be used once
- D) It cannot be passed to functions

86. What is duck typing (commonly associated with Python)?

- A) A type system that checks types strictly at compile time
- B) An object's suitability is determined by the presence of certain methods/behavior rather than its explicit type
- C) A type of exception
- D) A memory allocation strategy

87. What is the default value of an uninitialized boolean instance variable in Java?

- A) true
- B) false
- C) null
- D) 0

88. What is the difference between a shallow copy and a deep copy of a list containing nested lists?

- A) A shallow copy also copies nested objects recursively, deep copy does not
- B) A shallow copy copies references to nested objects, deep copy recursively copies nested objects too
- C) They behave identically
- D) Deep copy is not possible in most languages

89. What is the purpose of a 'static' variable in a class (Java/C++)?

- A) It belongs to the instance and is unique per object
- B) It belongs to the class rather than any instance and is shared across all instances
- C) It can only be used inside loops
- D) It is always private

90. Which of the following is true about Python's Global Interpreter Lock (GIL)?

- A) It allows true parallel execution of threads on multiple cores
- B) It ensures only one thread executes Python bytecode at a time in CPython, limiting CPU-bound multithreading performance
- C) It has no impact on multithreading
- D) It is used only in Python 2

91. Which of the following is a value type (not a reference type) in Java?

- A) int
- B) String
- C) Array
- D) Object

92. In C++, what does the 'const' keyword do when applied to a variable?

- A) Makes the variable a pointer
- B) Prevents the variable's value from being modified after initialization
- C) Declares a static variable
- D) Declares a global variable

93. What is method chaining?

- A) Calling multiple methods on the same object in a single statement by having each method return the object itself
- B) A way to declare multiple classes
- C) A type of exception handling
- D) A memory management technique

94. In Java, what is autoboxing?

- A) Automatic conversion between primitive types and their corresponding wrapper classes
- B) A way to box multiple objects together
- C) A garbage collection technique
- D) A type of casting error

Web & Cloud Fundamentals

95. What is the purpose of cookies in web applications?

- A) To store small pieces of data on the client side to maintain state/session information
- B) To encrypt HTTPS traffic
- C) To speed up DNS resolution
- D) To compress images

96. What is the difference between IaaS, PaaS, and SaaS?

- A) They are all the same cloud service model
- B) IaaS provides raw infrastructure, PaaS provides a platform to build apps, SaaS delivers ready-to-use software
- C) SaaS provides infrastructure while IaaS provides software
- D) PaaS is only for storage

97. Which HTTP method is typically used to retrieve data without side effects?

- A) POST
- B) GET
- C) DELETE
- D) PUT

98. What is a RESTful API?

- A) An API that strictly requires SOAP protocol
- B) An architectural style for designing networked applications using stateless operations over standard HTTP methods
- C) A type of database
- D) A programming language

99. What is auto-scaling in cloud computing?

- A) Automatically adjusting compute resources up or down based on demand
- B) Manually adding servers only
- C) A type of database indexing
- D) A billing model only

100. What does CSS stand for?

- A) Cascading Style Sheets
- B) Computer Style Sheets
- C) Creative Style System
- D) Cascading Software Sheets

Answer Key

1-B	2-A	3-B	4-A	5-B	6-A	7-A	8-B	9-A	10-A
11-A	12-A	13-A	14-A	15-B	16-A	17-A	18-A	19-B	20-B
21-A	22-B	23-B	24-B	25-B	26-B	27-A	28-B	29-A	30-C
31-B	32-A	33-C	34-A	35-A	36-B	37-D	38-B	39-B	40-A
41-A	42-A	43-A	44-A	45-B	46-B	47-A	48-B	49-A	50-C
51-A	52-A	53-A	54-A	55-B	56-C	57-A	58-B	59-A	60-C
61-A	62-A	63-B	64-C	65-C	66-C	67-A	68-B	69-A	70-B
71-B	72-B	73-B	74-B	75-C	76-B	77-A	78-C	79-B	80-B
81-B	82-A	83-C	84-A	85-B	86-B	87-B	88-B	89-B	90-B
91-A	92-B	93-A	94-A	95-A	96-B	97-B	98-B	99-A	100-A

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